MASTER FUNCTIONAL SKILLS

FUNCTIONAL SKILLS MATHEMATICS

ENTRY LEVEL 3 CORE GUIDE • VISUAL STEPS

ENTRY 3

FOUNDATIONS

MATHEMATICS
ENTRY LEVEL 3

Step-by-step visual numeracy foundations for Entry Level 3 candidates

Entry 3 Spec | Step-by-Step | Number & Money | Visual Models

MATHS ENTRY 3

Functional Skills Mathematics

Entry Level 3 • Core Foundations, Concrete Models & Practical Problem Solving

This book belongs to:	Exam date:	Target level:
_____	_____	_____

What is inside

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How to use this book

Read
Read the rule. Study the picture.

Try
Write your answers on the lines.

Check
Mark it with the answers at the back.

Remember
Learn the Golden Rule by heart.

NUMBER • ENTRY 3

1 Counting, Reading & Writing Numbers up to 1,000



Everyday example for Counting, Reading & Writing Numbers up to 1,000.

How to Do It • Key Rule

Numbers up to 1,000 have three place value columns: Hundreds (H), Tens (T), and Units (U).

In 468: 4 hundreds (400), 6 tens (60), 8 units (8). Words: four hundred and sixty-eight.

Step-by-Step Example

Write the number five hundred and twenty-four in digits.

1. 5 hundreds = 500
2. 2 tens = 20
3. 4 units = 4
4. Combined digits = **524**

Top Tip

Always write digits clearly in the correct column.

Remember!

Don't write 50024 — remember each column takes one digit!

TOPIC 1 • CANDIDATE WORKSPACE

P Counting, Reading & Writing Numbers up to 1,000: Practice

1 **Task 1 (Visual Model):** Write the number **four hundred and fifty-eight** in digits. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** A warehouse shelf holds 6 hundreds, 0 tens, and 2 units of bolts. Write this number in digits. **(2 marks)**

Show your working

Answer: _____

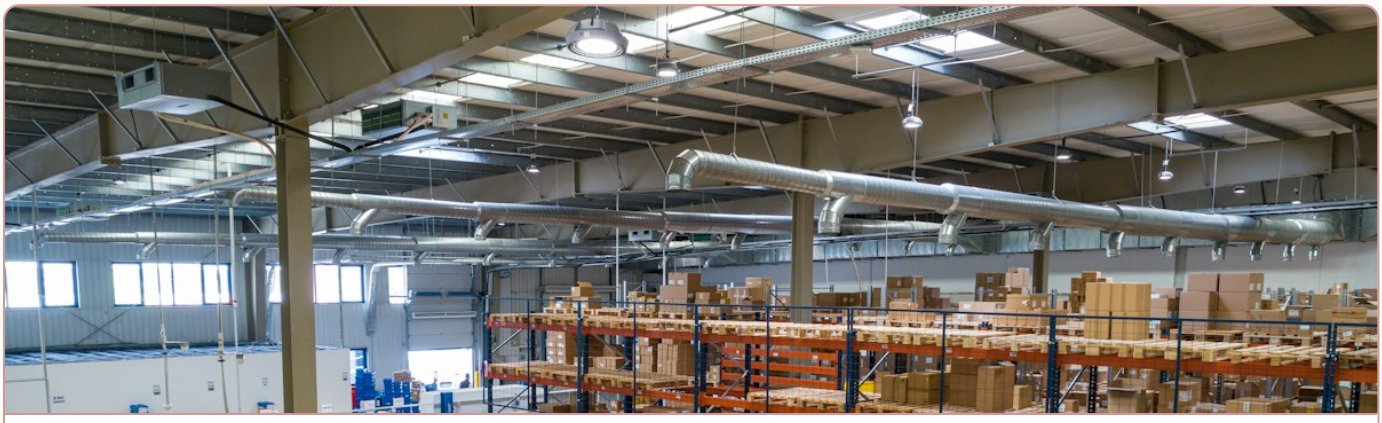
3 **Task 3 (Independent Challenge):** Put these numbers in order from smallest to largest: 420, 148, 680, 235. **(2 marks)**

Show your working

Answer: _____

NUMBER • ENTRY 3

2 Comparing & Ordering Numbers to 1,000



Everyday example for Comparing & Ordering Numbers to 1,000.

How to Do It • Key Rule

Compare numbers by looking at the largest place value column first (Hundreds, then Tens, then Units).

> means greater than (e.g. 520 > 480). < means less than (e.g. 310 < 390). = means equal to.

Step-by-Step Example

Put these parcel weights in order from lightest to heaviest: 482 g, 419 g, 503 g, 475 g.

1. Compare hundreds: 419, 475, 482 all start with 4; 503 starts with 5
2. Compare tens: 419 (1 ten), 475 (7 tens), 482 (8 tens)
3. Correct order: **419 g, 475 g, 482 g, 503 g**

Top Tip

The wide end of the > symbol always points towards the larger number.

Remember!

Always check whether question asks for smallest-to-largest or largest-to-smallest!

P

TOPIC 2 • CANDIDATE WORKSPACE

Comparing & Ordering Numbers to 1,000: Practice

1 **Task 1 (Visual Model):** Insert the correct symbol (**>** or **<**) to compare: 564 ___ 546. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** From these parcel weights (240 g, 125 g, 480 g, 315 g), write down the lowest weight and the highest weight. **(2 marks)**

Show your working

Answer: _____

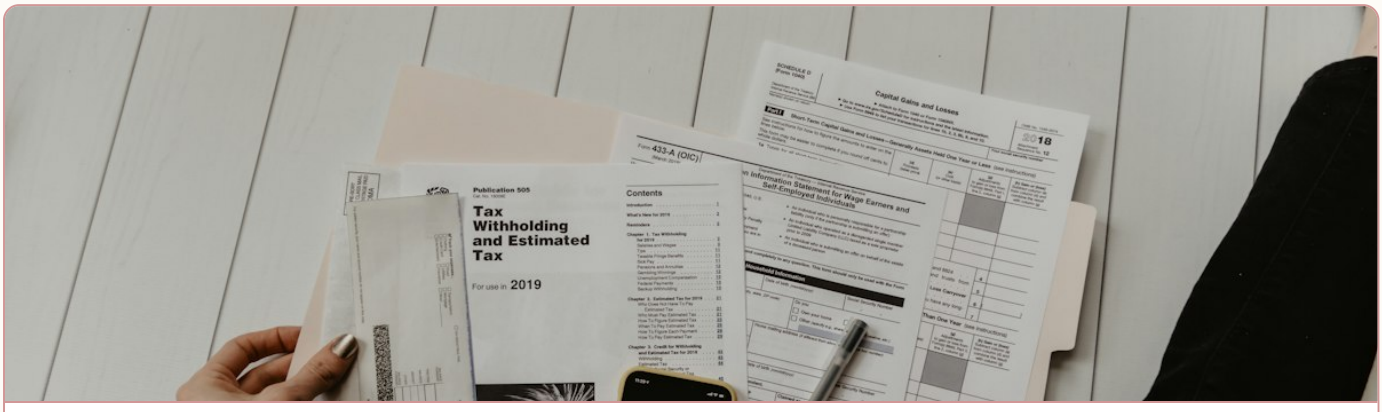
3 **Task 3 (Independent Challenge):** Put the four parcel weights above in order from lightest to heaviest. **(2 marks)**

Show your working

Answer: _____

NUMBER • ENTRY 3

3 Addition: Column Method with Regrouping



Everyday example for Addition: Column Method with Regrouping.

How to Do It • Key Rule

Column addition adds numbers in vertical columns from right to left (Units first, then Tens, then Hundreds).

If a column adds up to 10 or more, carry the 1 into the next column on the left.

Step-by-Step Example

A shop assistant sells 248 items on Friday and 375 items on Saturday. Find the total items sold.

1. Units: 8 + 5 = 13 (write 3, carry 1)
2. Tens: 4 + 7 + 1 = 12 (write 2, carry 1)
3. Hundreds: 2 + 3 + 1 = 6
4. Total = **623 items**

Top Tip

Always line up your columns from the right-hand side.

Remember!

Don't forget to add in any carried digits!

TOPIC 3 • CANDIDATE WORKSPACE

P

Addition: Column Method with Regrouping: Practice

1 **Task 1 (Visual Model):** Use column addition to work out: 245 + 138. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** You buy a coat for £34.50 and shoes for £18.20. Calculate the total cost. **(2 marks)**

Show your working

Answer: _____

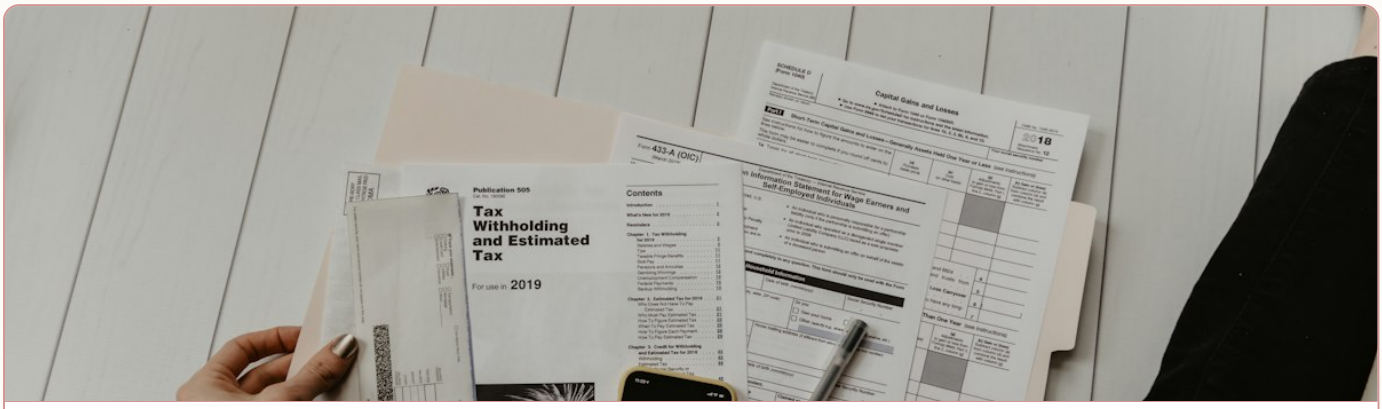
3 **Task 3 (Independent Challenge):** A clinic sees 124 patients on Monday and 98 patients on Tuesday. How many patients were seen in total? **(2 marks)**

Show your working

Answer: _____

NUMBER • ENTRY 3

4 Subtraction: Column Method with Borrowing



Everyday example for Subtraction: Column Method with Borrowing.

How to Do It • Key Rule

Column subtraction subtracts the bottom digit from the top digit in each column from right to left.

If the top digit is smaller than the bottom digit, borrow 1 from the column to the left.

Step-by-Step Example

A warehouse has 642 boxes in stock. 278 boxes are dispatched on a lorry. How many boxes remain?

1. Units: 2 - 8 cannot do; borrow 1 from tens -> 12 - 8 = 4
2. Tens: 3 - 7 cannot do; borrow 1 from hundreds -> 13 - 7 = 6
3. Hundreds: 5 - 2 = 3
4. Remaining = **364 boxes**

Top Tip

Never swap the digits around: if the top is smaller, you MUST borrow!

Remember!

Check your answer: 364 + 278 = 642.

P

TOPIC 4 • CANDIDATE WORKSPACE

Subtraction: Column Method with Borrowing: Practice

1 **Task 1 (Visual Model):** Use column subtraction with borrowing to work out: 642 – 278. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** A store room has 450 boxes of paper. Staff use 185 boxes. How many boxes are left? **(2 marks)**

Show your working

Answer: _____

3 **Task 3 (Independent Challenge):** Find the difference between 512 and 349. Show a check by adding. **(2 marks)**

Show your working

Answer: _____

NUMBER • ENTRY 3

5 Multiplication Facts: 2, 3, 4, 5 & 10



Everyday example for Multiplication Facts: 2, 3, 4, 5 & 10.

How to Do It • Key Rule

Multiplication is repeated addition. Learn your times tables to calculate quickly in everyday life.

2x table doubles numbers; 5x table ends in 0 or 5; 10x table adds a place-value zero.

Step-by-Step Example

A pharmacy orders 8 packs of face masks. Each pack contains 5 masks. How many masks altogether?

1. Calculation: 8 packs x 5 masks
2. Use 5 times table: 5, 10, 15, 20, 25, 30, 35, 40
3. Total masks = **40 masks**

Top Tip

Multiplication can be done in any order: 8 x 5 is the same as 5 x 8.

Remember!

Multiplying by 10 shifts digits one place to the left.

TOPIC 5 • CANDIDATE WORKSPACE

P

Multiplication Facts: 2, 3, 4, 5 & 10: Practice

1 **Task 1 (Visual Model):** Work out: 8 × 5. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** There are 6 tables in a training room with 4 name badges on each table. How many badges are there in total? **(2 marks)**

Show your working

Answer: _____

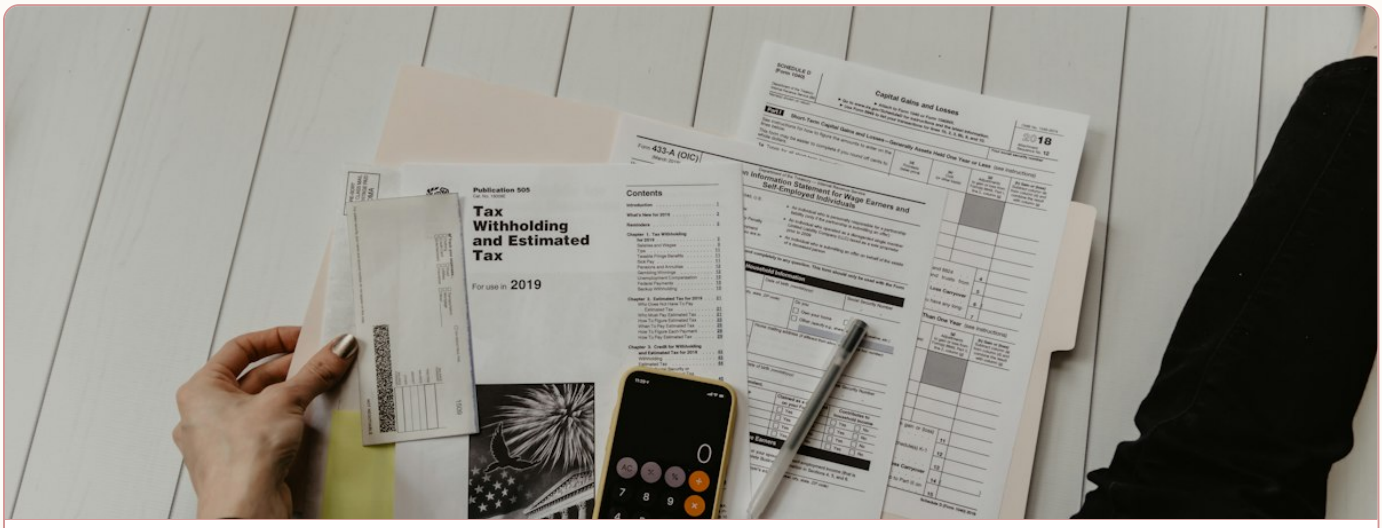
3 **Task 3 (Independent Challenge):** A care home orders 7 packs of tissues with 10 boxes in each pack. How many boxes is that altogether? **(2 marks)**

Show your working

Answer: _____

NUMBER • ENTRY 3

6 Multiplying 2-Digit Numbers by 1-Digit



Everyday example for Multiplying 2-Digit Numbers by 1-Digit.

How to Do It • Key Rule

To multiply a 2-digit number by a single digit: Multiply the units first, then multiply the tens.

Partitioning method: 24 x 3 = (20 x 3) + (4 x 3) = 60 + 12 = 72.

Step-by-Step Example

A baker makes 3 batches of muffins. Each batch has 24 muffins. How many muffins in total?

1. Multiply units: 4 x 3 = 12 (write 2, carry 1)
2. Multiply tens: 2 x 3 = 6; add carry: 6 + 1 = 7
3. Total muffins = **72 muffins**

Top Tip

Break large numbers into tens and units to multiply mentally.

Remember!

Check: 25 x 3 = 75, so 72 is very sensible.

P

TOPIC 6 • CANDIDATE WORKSPACE

Multiplying 2-Digit Numbers by 1-Digit: Practice

1 **Task 1 (Visual Model):** Multiply: 24 × 3. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** A delivery consists of 4 crates, each holding 35 cartons of juice. How many cartons are delivered? **(2 marks)**

Show your working

Answer: _____

3 **Task 3 (Independent Challenge):** Work out 42 × 5, showing your working clearly. **(2 marks)**

Show your working

Answer: _____

NUMBER • ENTRY 3

7 Division Facts & Sharing with Remainders



Everyday example for Division Facts & Sharing with Remainders.

How to Do It • Key Rule

Division shares an amount into equal groups: Total / Number of groups = Items per group.

Any leftover amount is called the remainder.

Step-by-Step Example

Share 29 safety badges equally between 4 team members. How many does each get, and how many are left over?

1. 4 x 7 = 28
2. 29 - 28 = 1 remainder
3. Each person gets **7 badges** with **1 left over**

Top Tip

Multiplication is the opposite of division: 28 / 4 = 7 because 7 x 4 = 28.

Remember!

Remainder must always be smaller than the number you divided by.

P

TOPIC 7 • CANDIDATE WORKSPACE

Division Facts & Sharing with Remainders: Practice

1 **Task 1 (Visual Model):** Divide: 28 ÷ 4. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** Share 29 safety badges equally between 4 workers. How many does each worker get, and how many are left over? **(2 marks)**

Show your working

Answer: _____

3 **Task 3 (Independent Challenge):** 45 apprentices are put into equal teams of 5. How many teams are there? **(2 marks)**

Show your working

Answer: _____

NUMBER • ENTRY 3

8 Fractions of Shapes & Amounts: 1/2, 1/4, 1/3, 1/10



Everyday example for Fractions of Shapes & Amounts: 1/2, 1/4, 1/3, 1/10.

How to Do It • Key Rule

A fraction represents equal parts of a whole. 1/2 = half (divide by 2); 1/4 = quarter (divide by 4).

1/3 = third (divide by 3); 1/10 = tenth (divide by 10).

Step-by-Step Example

Find 1/4 of £32 for a customer discount.

1. Divide by 4: £32 / 4
2. Calculation: 4 x 8 = 32
3. Discount = **£8.00**

Top Tip

To find half, divide by 2. To find a quarter, halve and halve again.

Remember!

All parts of a fraction MUST be equal in size.

TOPIC 8 • CANDIDATE WORKSPACE

P Fractions of Shapes & Amounts: 1/2, 1/4, 1/3, 1/10: Practice

1 **Task 1 (Visual Model):** Find 1/2 (half) of £40. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** Cut a 32-metre rope into 4 equal quarters. How long is 1/4 of the rope? **(2 marks)**

Show your working

Answer: _____

3 **Task 3 (Independent Challenge):** Find 1/10 (one-tenth) of 100 kg of potatoes. **(2 marks)**

Show your working

Answer: _____

NUMBER • ENTRY 3

9 Decimals: Tenths, Money & Place Value



Everyday example for Decimals: Tenths, Money & Place Value.

How to Do It • Key Rule

Decimals use a dot to separate whole numbers from fractional parts. Money uses 2 decimal places (£ and pence).

£4.50 = 4 whole pounds and 50 pence (5 tenths of a pound).

Step-by-Step Example

Write seventy-five pence in decimal pound notation.

1. There are 0 whole pounds
2. 75 pence is 75/100 of a pound
3. Decimal notation = **£0.75**

Top Tip

Always write two digits after the decimal point for money (£3.50, not £3.5).

Remember!

Remember: 100 pence = 1 pound.

TOPIC 9 • CANDIDATE WORKSPACE

P

Decimals: Tenths, Money & Place Value: Practice

1 **Task 1 (Visual Model):** Write seventy-five pence in decimal pound notation (£). **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** Add together £4.50 and £2.25. **(2 marks)**

Show your working

Answer: _____

3 **Task 3 (Independent Challenge):** Subtract £6.50 from £10.00. **(2 marks)**

Show your working

Answer: _____

10

NUMBER • ENTRY 3

Rounding to Nearest 10 & 100



Everyday example for Rounding to Nearest 10 & 100.

How to Do It • Key Rule

Rounding gives a simple approximate number that is close to the real value.

Rounding rule: Look at the next digit to the right. 5 or more rounds up; 4 or less stays the same.

Step-by-Step Example

Round 67 to the nearest 10 for a quick stock estimate.

1. Units digit is 7
2. Since 7 >= 5, round the tens digit up from 60 to 70
3. Rounded = **70**

Top Tip

Numbers ending in 5 always round UP to the higher ten.

Remember!

Rounding helps check your answers are sensible.

TOPIC 10 • CANDIDATE WORKSPACE

P

Rounding to Nearest 10 & 100: Practice

1 **Task 1 (Visual Model):** Round 67 to the nearest 10. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** Round 235 to the nearest 100. **(2 marks)**

Show your working

Answer: _____

3 **Task 3 (Independent Challenge):** Round 85 to the nearest 10 and explain which rule you used. **(2 marks)**

Show your working

Answer: _____

MEASURE • ENTRY 3

11 Money Calculations: Adding Costs & Giving Change



Everyday example for Money Calculations: Adding Costs & Giving Change.

How to Do It • Key Rule

Add prices to find the total bill. Subtract from the payment note to calculate change.

Change = Payment note - Total cost.

Step-by-Step Example

You buy a sandwich for £2.85 and a hot drink for £1.65. You pay with a £5 note. Find your change.

1. Total cost: £2.85 + £1.65 = £4.50
2. Subtract from £5.00: £5.00 - £4.50
3. Change = **£0.50** (50p)

Top Tip

Count up from £4.50 to £5.00: add 50p to make £5.

Remember!

Always check you haven't dropped any pence digits.

TOPIC 11 • CANDIDATE WORKSPACE

P Money Calculations: Adding Costs & Giving Change: Practice

1 **Task 1 (Visual Model):** You buy a sandwich for £2.85 and a drink for £1.65, and pay with a £5 note. How much change do you get? **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** Calculate the change from a £10 note when spending £6.40. **(2 marks)**

Show your working

Answer: _____

3 **Task 3 (Independent Challenge):** You pay for a £14.50 bus pass with a £20 note. Work out your change. **(2 marks)**

Show your working

Answer: _____

MEASURE • ENTRY 3

12 Telling Time: Clock Faces & Digital 12h/24h



Everyday example for Telling Time: Clock Faces & Digital 12h/24h.

How to Do It • Key Rule

Clock hands: Short hand points to hours; long hand points to minutes (each number = 5 minutes).

Digital time shows hours : minutes. 24h clock: 14:00 is 2:00 pm (add 12 to afternoon hours).

Step-by-Step Example

A shift starts at quarter past 9 in the morning. Write this time in digital 12-hour format.

1. Quarter past is 15 minutes past the hour
2. Morning is am
3. Digital time = **9:15 am**

Top Tip

60 minutes = 1 hour; 30 minutes = half past; 15 minutes = quarter past.

Remember!

Afternoon times in 24h add 12 to the hour.

P

TOPIC 12 • CANDIDATE WORKSPACE

Telling Time: Clock Faces & Digital 12h/24h: Practice

1 **Task 1 (Visual Model):** Write 'quarter past 9 in the morning' in 12-hour digital format. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** Write 2:00 pm in 24-hour digital clock format. **(2 marks)**

Show your working

Answer: _____

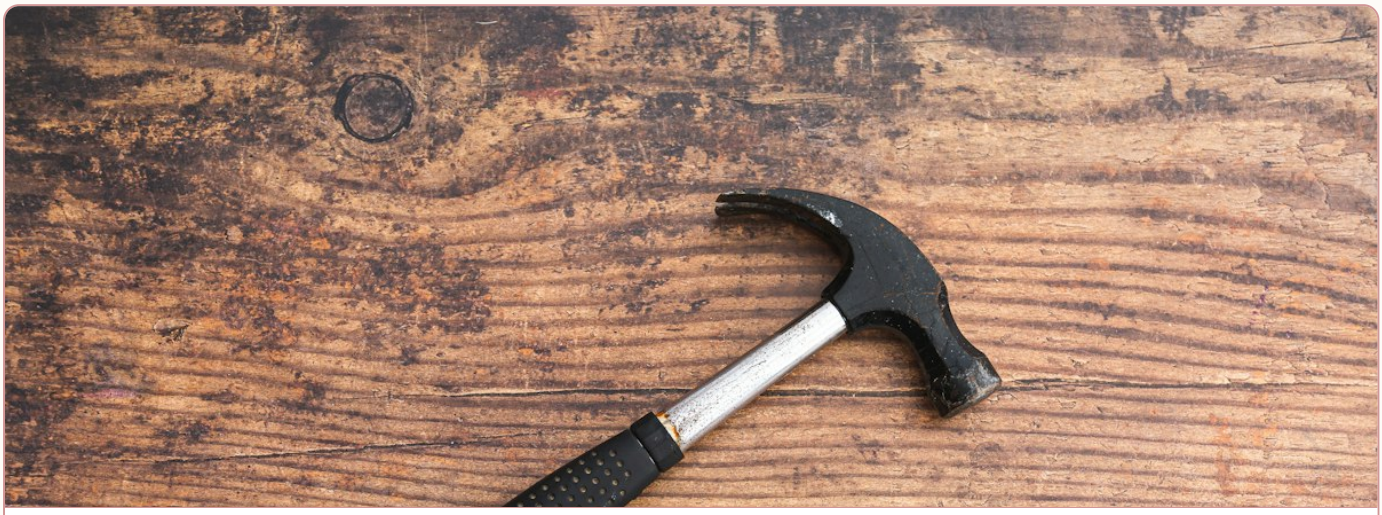
3 **Task 3 (Independent Challenge):** Write 'half past 4 in the afternoon' in 24-hour clock format. **(2 marks)**

Show your working

Answer: _____

MEASURE • ENTRY 3

13 Measuring Length: mm, cm and metres



Everyday example for Measuring Length: mm, cm and metres.

How to Do It • Key Rule

Use a ruler or tape measure to measure lengths accurately in metric units.

10 millimetres (mm) = 1 centimetre (cm). 100 centimetres (cm) = 1 metre (m).

Step-by-Step Example

A table is 1 metre and 40 cm long. What is its total length in centimetres?

1. 1 metre = 100 cm
2. Add 40 cm: 100 + 40
3. Total length = **140 cm**

Top Tip

Start measuring from the 0 mark on the ruler, NOT from the wooden end!

Remember!

Small objects use mm or cm; rooms and buildings use metres.

TOPIC 13 • CANDIDATE WORKSPACE

P

Measuring Length: mm, cm and metres: Practice

1 **Task 1 (Visual Model):** A workbench is 1 metre and 40 cm long. What is this length in centimetres? **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** Convert 25 millimetres (mm) into centimetres (cm). **(2 marks)**

Show your working

Answer: _____

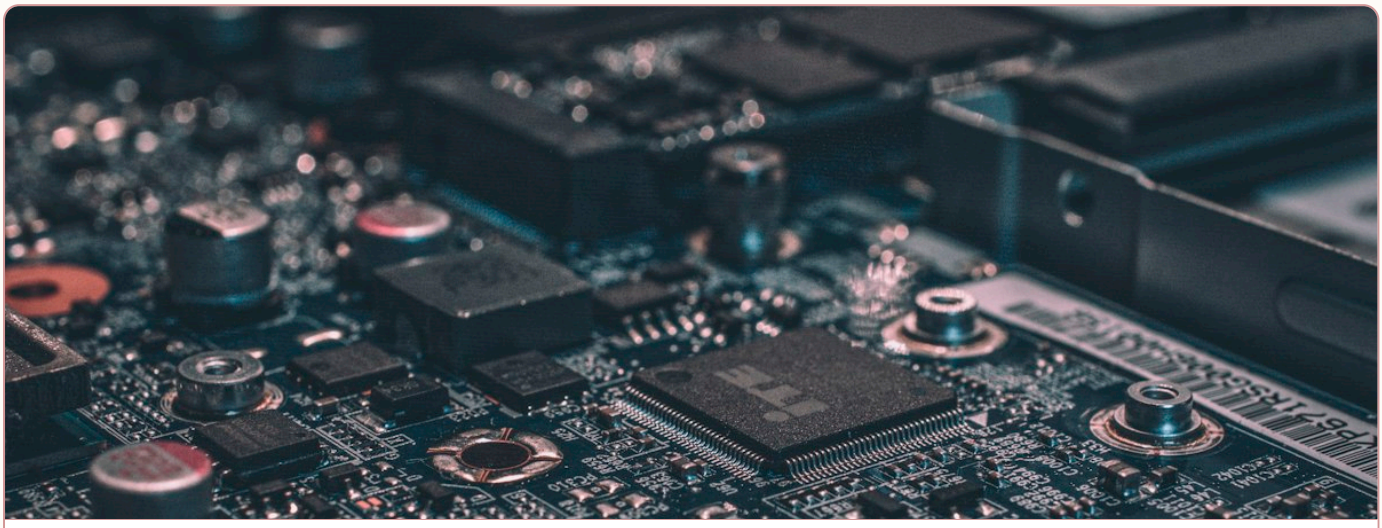
3 **Task 3 (Independent Challenge):** Convert 3.2 metres into centimetres. **(2 marks)**

Show your working

Answer: _____

MEASURE • ENTRY 3

14 Measuring Weight: grams (g) and kilograms (kg)



Everyday example for Measuring Weight: grams (g) and kilograms (kg).

How to Do It • Key Rule

Weighing scales measure how heavy objects are. 1,000 grams (g) = 1 kilogram (kg).

1 kg is roughly the weight of a standard bag of sugar.

Step-by-Step Example

A recipe needs half a kilogram of flour. How many grams is this?

1. 1 kg = 1,000 g
2. Half of 1,000 g: 1,000 / 2
3. Flour required = **500 grams**

Top Tip

Always check the scale dial increments: each tick mark might be 10g, 50g, or 100g.

Remember!

Letters and food ingredients use grams; luggage and people use kg.

TOPIC 14 • CANDIDATE WORKSPACE

P

Measuring Weight: grams (g) and kilograms (kg): Practice

1 **Task 1 (Visual Model):** How many grams are in half a kilogram (1/2 kg)? **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** Convert 1.5 kilograms of apples into grams. **(2 marks)**

Show your working

Answer: _____

3 **Task 3 (Independent Challenge):** You pack a 750 g book and a 250 g notebook into a bag. What is their total weight in kilograms? **(2 marks)**

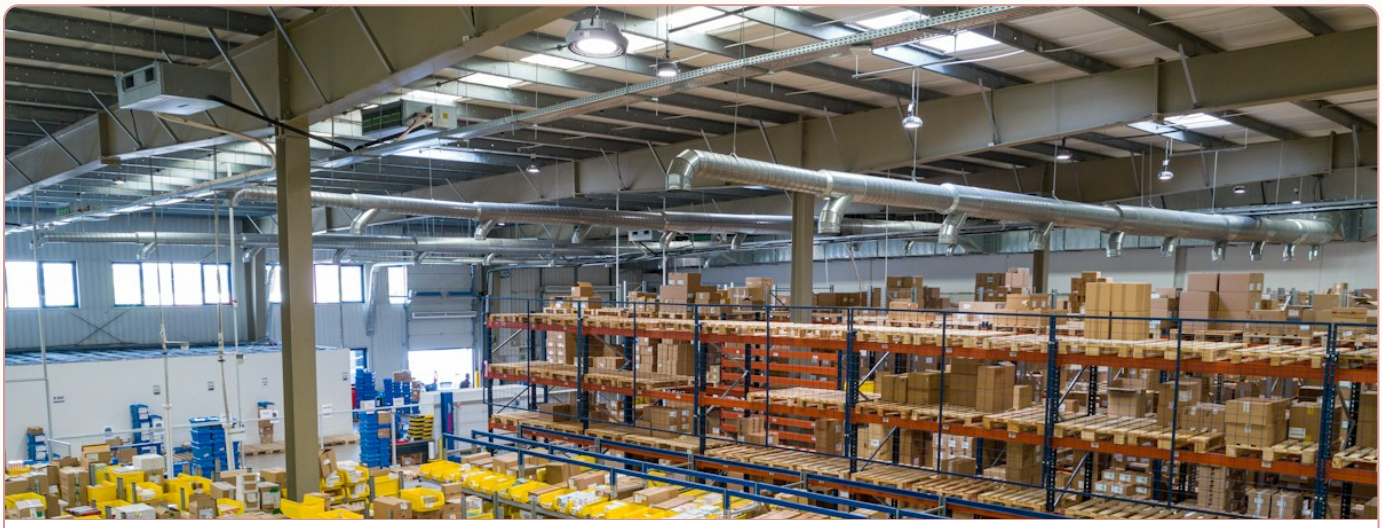
Show your working

Answer: _____

MEASURE • ENTRY 3

15

Measuring Capacity: millilitres (ml) and litres (l)



Everyday example for Measuring Capacity: millilitres (ml) and litres (l).

How to Do It • Key Rule

Measuring jugs measure liquids in millilitres (ml) and litres (l).

1,000 ml = 1 litre.

Half a litre = 500 ml. Quarter of a litre = 250 ml.

Step-by-Step Example

A kettle holds 1.5 litres of water. Convert this capacity into millilitres.

1. 1 litre = 1,000 ml
2. 0.5 litre = 500 ml
3. Total capacity = **1,500 ml**

Top Tip

Look at the water level from eye level at the bottom of the curved line (meniscus).

Remember!

Medicine spoons hold 5 ml.

TOPIC 15 • CANDIDATE WORKSPACE

P Measuring Capacity: millilitres (ml) and litres (l): Practice

1 **Task 1 (Visual Model):** A kettle holds 1.5 litres of water. How many millilitres (ml) is this? **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** How many millilitres are in half a litre? **(2 marks)**

Show your working

Answer: _____

3 **Task 3 (Independent Challenge):** Convert 2,000 millilitres into litres. **(2 marks)**

Show your working

Answer: _____

MEASURE • ENTRY 3

16 Reading Temperature Scales in °C



Everyday example for Reading Temperature Scales in °C.

How to Do It • Key Rule

Thermometers measure temperature in degrees Celsius (°C). Zero (0°C) is the freezing point of water.

Temperatures above zero are warm; temperatures below zero (negative) are freezing cold.

Step-by-Step Example

The thermometer in a medical room reads 21°C. The recommended temperature is 18°C. By how much is it too warm?

1. Subtract: 21°C - 18°C
2. Difference = **3°C** too warm

Top Tip

Look at the markings on the thermometer to see if each line is 1°C or 2°C.

Remember!

Room temperature is usually around 20°C.

P

TOPIC 16 • CANDIDATE WORKSPACE

Reading Temperature Scales in °C: Practice

1 **Task 1 (Visual Model):** A room thermometer reads 21°C, and the target temperature is 18°C. By how many degrees is it too warm? **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** What is the freezing point of water in degrees Celsius (°C)? **(2 marks)**

Show your working

Answer: _____

3 **Task 3 (Independent Challenge):** Outside temperature rises from −2°C in the morning to +5°C at midday. By how many degrees did it rise? **(2 marks)**

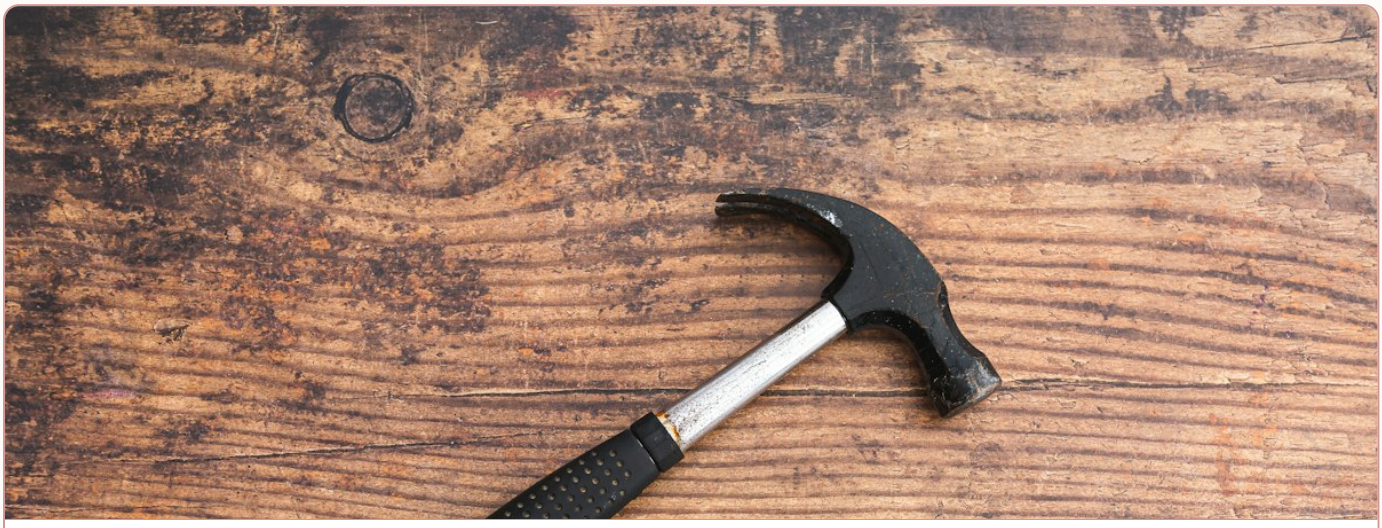
Show your working

Answer: _____

MEASURE • ENTRY 3

17

Perimeter: Distance Around a 2D Shape



Everyday example for Perimeter: Distance Around a 2D Shape.

How to Do It • Key Rule

Perimeter is the total distance all the way around the outside edges of a flat shape.

To find the perimeter: Add the lengths of all the sides together.

Step-by-Step Example

A rectangular notice board is 80 cm long and 50 cm wide. Find its perimeter.

1. Four sides: 80 cm + 50 cm + 80 cm + 50 cm
2. Calculation: 160 + 100
3. Perimeter = **260 cm** (or 2.6 m)

Top Tip

Remember to add all four sides of a rectangle, not just the two labelled ones!

Remember!

Tick off each side as you add it.

P

TOPIC 17 • CANDIDATE WORKSPACE

Perimeter: Distance Around a 2D Shape: Practice

1 **Task 1 (Visual Model):** A rectangular notice board is 80 cm long and 50 cm wide. Calculate its perimeter. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** A rectangular flowerbed measures 6 metres by 4 metres. Find its total perimeter. **(2 marks)**

Show your working

Answer: _____

3 **Task 3 (Independent Challenge):** A square patio has sides of 5 metres. What is the perimeter of the square? **(2 marks)**

Show your working

Answer: _____

MEASURE • ENTRY 3

18 Recognising 2D & 3D Shapes in Daily Life



Everyday example for Recognising 2D & 3D Shapes in Daily Life.

How to Do It • Key Rule

2D flat shapes: triangle (3 sides), square (4 equal sides), rectangle (4 sides), circle (round).

3D solid shapes: cube (box with square faces), cuboid (box with rectangle faces), cylinder (tube/tin can), sphere (ball).

Step-by-Step Example

A food tin has two circular flat ends and one curved side. Name this 3D shape.

1. Two circular faces and curved surface = **Cylinder**
2. Examples: soup tin, coin roll, water pipe

Top Tip

2D shapes are flat on paper; 3D shapes are solid objects you can pick up.

Remember!

A square is a special type of rectangle where all 4 sides are equal.

P

TOPIC 18 • CANDIDATE WORKSPACE

Recognising 2D & 3D Shapes in Daily Life: Practice

1 **Task 1 (Visual Model):** Name the 3D shape of a standard soup tin with two circular ends. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** Name the 2D flat shape that has 4 equal sides and 4 right-angle corners. **(2 marks)**

Show your working

Answer: _____

3 **Task 3 (Independent Challenge):** Name the 3D shape of a cereal box that has 6 rectangular faces. **(2 marks)**

Show your working

Answer: _____

DATA • ENTRY 3

19 Position, Movement & Compass Directions



Everyday example for Position, Movement & Compass Directions.

How to Do It • Key Rule

Use position words: left, right, forwards, backwards, between, next to, opposite.

Four compass points: North, East, South, West (clockwise from top).

Step-by-Step Example

You walk North along a corridor, then turn right. Which compass direction are you now walking in?

1. Facing North (up)
2. Turning right is 90° clockwise
3. New direction = **East**

Top Tip

Remember: 'Never Eat Shredded Wheat' for North, East, South, West.

Remember!

A quarter turn clockwise is turning to the right.

TOPIC 19 • CANDIDATE WORKSPACE

P

Position, Movement & Compass Directions: Practice

1 **Task 1 (Visual Model):** You are facing North and make a quarter turn to the right (clockwise). Which direction are you now facing? **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** Which compass direction is directly opposite North? **(2 marks)**

Show your working

Answer: _____

3 **Task 3 (Independent Challenge):** You are facing East and turn 90° left (anti-clockwise). Which direction are you facing? **(2 marks)**

Show your working

Answer: _____

DATA • ENTRY 3

20 Reading Information from Grid References



Everyday example for Reading Information from Grid References.

How to Do It • Key Rule

Grid references pinpoint locations on a map or floor plan using letters and numbers.

Always read the horizontal coordinate first, then the vertical coordinate: 'Along the corridor and up the stairs'.

Step-by-Step Example

On a hospital ward map, the Pharmacy is in column C, row 4. What is its grid reference?

1. Letter first: C
2. Number second: 4
3. Grid reference = **C4**

Top Tip

Always write the horizontal letter first, followed by the vertical number.

Remember!

Check your finger points to the square where row and column meet.

TOPIC 20 • CANDIDATE WORKSPACE

P

Reading Information from Grid References: Practice

1 **Task 1 (Visual Model):** On a floor plan grid, the First Aid room is in Column C, Row 4. Write its grid reference. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** The Pharmacy is in Column B, Row 2. Write its grid reference. **(2 marks)**

Show your working

Answer: _____

3 **Task 3 (Independent Challenge):** Reception is in Column A, Row 1. Write its grid reference. **(2 marks)**

Show your working

Answer: _____

DATA • ENTRY 3

21 Collecting Data: Tally Charts & Lists



Everyday example for Collecting Data: Tally Charts & Lists.

How to Do It • Key Rule

Tally marks count items in groups of five: four vertical lines and one diagonal strike.

Count tallies by fives to find totals quickly and accurately.

Step-by-Step Example

A café records drinks sold during breakfast: Coffee has 3 full tally bundles and 2 single lines. Find total coffees.

1. Each bundle of 5 = 5 tallies
2. 3 bundles: 3 x 5 = 15
3. Add 2 single lines: 15 + 2 = **17 coffees**

Top Tip

Count in fives first: 5, 10, 15, then count on: 16, 17.

Remember!

Draw the diagonal line through the group on the fifth count.

P

TOPIC 21 • CANDIDATE WORKSPACE

Collecting Data: Tally Charts & Lists: Practice

1 **Task 1 (Visual Model):** A tally chart shows 3 bundles of five and 2 single lines for coffee sales. How many coffees were sold? **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** Draw or describe how to write 8 in tally marks. **(2 marks)**

Show your working

Answer: _____

3 **Task 3 (Independent Challenge):** A cafe sold 17 coffees, 9 teas, and 6 hot chocolates. What is the total tally count of hot drinks? **(2 marks)**

Show your working

Answer: _____

DATA • ENTRY 3

22 Reading & Comparing Data in Bar Charts



Everyday example for Reading & Comparing Data in Bar Charts.

How to Do It • Key Rule

Bar charts use rectangular bars of different heights to show numbers for different categories.

Read the height of each bar against the scale on the left side.

Step-by-Step Example

In a college survey bar chart, 18 learners study English, 24 study Maths, and 12 study Digital Skills. Which subject has the most learners?

1. Compare bar heights: English = 18, Maths = 24, Digital = 12
2. The tallest bar is Maths with **24 learners**

Top Tip

Look at the scale numbers carefully: check if each line is 1, 2, 5, or 10.

Remember!

Use a straight ruler across from the top of the bar to the number scale.

TOPIC 22 • CANDIDATE WORKSPACE

P

Reading & Comparing Data in Bar Charts: Practice

1 **Task 1 (Visual Model):** In a bar chart, 18 learners study English, 24 study Maths, and 12 study Digital Skills. Which subject has the tallest bar, and how many learners study it? **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** From the bar chart in Task 1, how many learners study English? **(2 marks)**

Show your working

Answer: _____

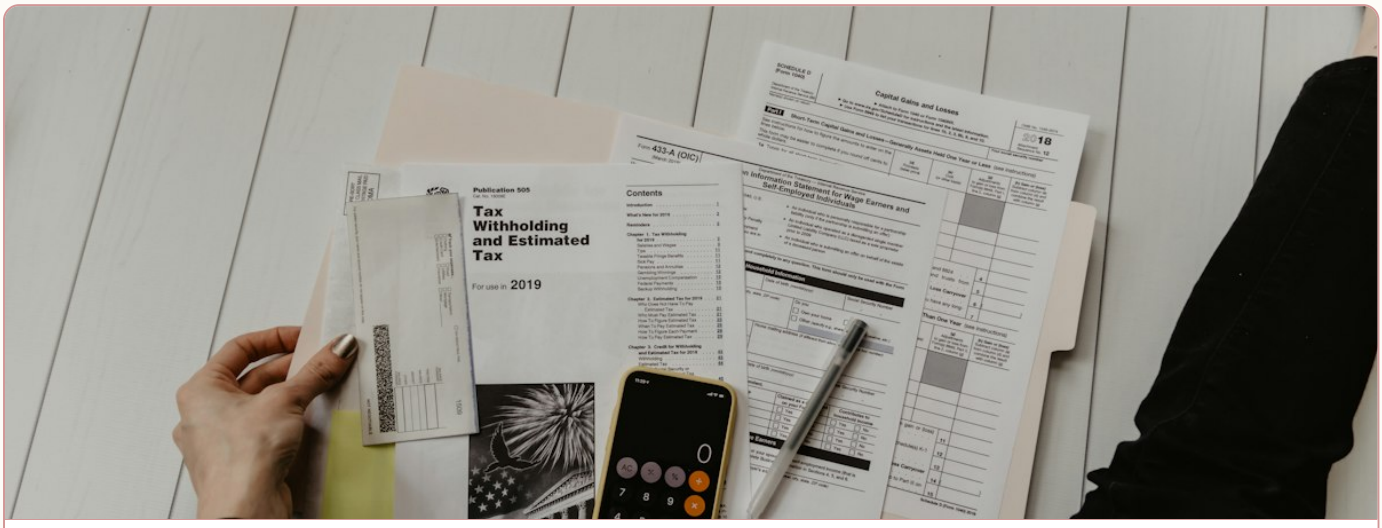
3 **Task 3 (Independent Challenge):** How many more learners study Maths than Digital Skills? **(2 marks)**

Show your working

Answer: _____

DATA • ENTRY 3

23 Sorting Data Using Venn & Carroll Diagrams



Everyday example for Sorting Data Using Venn & Carroll Diagrams.

How to Do It • Key Rule

Venn diagrams sort items into overlapping circles. The overlap area contains items with BOTH properties.

Carroll diagrams sort items into a 2x2 grid based on whether they have or do not have certain features.

Step-by-Step Example

In a health check of 20 workers, 12 wear glasses and 8 do not. Where do the 8 go in a Carroll table?

1. Under the column labelled 'Does NOT wear glasses'
2. Count = **8 workers**

Top Tip

Items outside the circles do not belong to either category.

Remember!

Check that all sorted items add up to the total starting number.

P

TOPIC 23 • CANDIDATE WORKSPACE

Sorting Data Using Venn & Carroll Diagrams: Practice

1 **Task 1 (Visual Model):** In a survey of 20 workers, 12 wear glasses and 8 do not. In which column of a Carroll diagram do the 8 workers go? **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** In a Venn diagram with two overlapping circles, what does the middle overlapping section show? **(2 marks)**

Show your working

Answer: _____

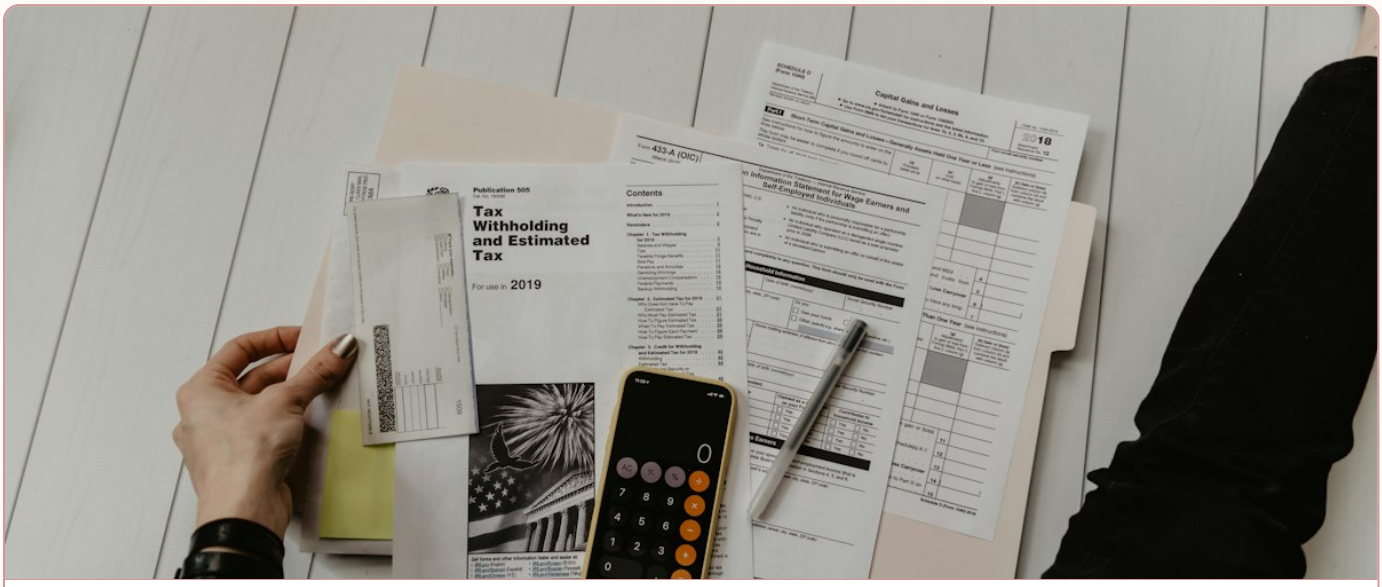
3 **Task 3 (Independent Challenge):** Check that the two groups in Task 1 (12 and 8) add up to the total number of workers surveyed. **(2 marks)**

Show your working

Answer: _____

DATA • ENTRY 3

24 Probability Words: Certain, Likely, Unlikely, Impossible



Everyday example for Probability Words: Certain, Likely, Unlikely, Impossible.

How to Do It • Key Rule

Use everyday words to describe how likely an event is to happen.

Certain: guaranteed to happen. Likely: probably will happen. Unlikely: probably will not happen. Impossible: cannot happen.

Step-by-Step Example

Describe the likelihood that the sun will rise tomorrow morning.

1. Guaranteed natural law
2. Likelihood = **Certain**

Top Tip

Impossible means zero chance (e.g. rolling an 8 on a standard 6-sided dice).

Remember!

Evens means equal 50-50 chance (e.g. flipping heads on a coin).

TOPIC 24 • CANDIDATE WORKSPACE

P

Probability Words: Certain, Likely, Unlikely, Impossible: Practice

1 **Task 1 (Visual Model):** Use a probability word (Certain, Likely, Unlikely, Impossible) to describe: 'The sun will rise tomorrow morning'. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Everyday Context):** Describe the likelihood that it will snow in London in July. **(2 marks)**

Show your working

Answer: _____

3 **Task 3 (Independent Challenge):** Describe the likelihood of rolling an 8 on a standard 6-sided dice numbered 1 to 6. **(2 marks)**

Show your working

Answer: _____

VOCATIONAL CALCULATION WORKSHOP • WORKSHOP 1

W1 Workshop 1: Retail Cashier & Till Operations

Workplace Scenario

Adding purchases, calculating 10% discounts, and giving change.

Use your maths skills to complete the workplace tasks below:

1 **Task 1 (Retail Transaction):** A customer buys bread for £1.45, butter for £2.10, and jam for £1.80. Calculate the total cost. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Change Given):** The customer pays with a £10 note. How much change does the cashier hand back? **(2 marks)**

Show your working

Answer: _____

CANDIDATE WORKSPACE • WORKSHOP 1

Workshop 1 • Retail Cashier & Till Operations Working Space

▶ Workspace Instructions • Retail Cashier & Till Operations

Use this workspace to lay out calculation steps or diagram sketches for **Retail Cashier & Till Operations**. Record currency symbols (£, p) clearly with all change calculations.

✎ Candidate Working Grid • Retail Cashier & Till Operations

VOCATIONAL CALCULATION WORKSHOP • WORKSHOP 2

W2

Workshop 2: Care Home Resident Medication & Liquids

Workplace Scenario

Measuring millilitres, tracking daily fluids, and reading thermometer scales.

Use your maths skills to complete the workplace tasks below:

1 **Task 1 (Daily Fluid Target):** A care home resident must drink 1,500 ml of fluids daily. He drinks 350 ml of tea, 400 ml of water, and 250 ml of juice. How many ml has he drunk? **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Remaining Fluid):** How many more millilitres does he need to drink to meet his daily target? **(2 marks)**

Show your working

Answer: _____

CANDIDATE WORKSPACE • WORKSHOP 2

Workshop 2 • Care Home Resident Medication & Liquids Working Space

Workspace Instructions • Care Home Resident Medication & Liquids

Use this workspace to lay out calculation steps or diagram sketches for **Care Home Resident Medication & Liquids**. Check that fluid measurements clearly state millilitres (ml) or litres (l).

✎ Candidate Working Grid • Care Home Resident Medication & Liquids

VOCATIONAL CALCULATION WORKSHOP • WORKSHOP 3

W3

Workshop 3: Catering Kitchen Recipe Scaling

Workplace Scenario

Doubling muffin batches, weighing flour in grams, and timing oven baking.

Use your maths skills to complete the workplace tasks below:

1 **Task 1 (Recipe Scaling):** A muffin recipe making 12 muffins needs 250 g of flour and 100 g of sugar. Calculate the amount of flour needed to make 24 muffins. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Baking Time):** Muffins go into the oven at 10:15 and bake for 25 minutes. At what time must they be taken out? **(2 marks)**

Show your working

Answer: _____

CANDIDATE WORKSPACE • WORKSHOP 3

Workshop 3 • Catering Kitchen Recipe Scaling Working Space

Workspace Instructions • Catering Kitchen Recipe Scaling

Use this workspace to lay out calculation steps or diagram sketches for **Catering Kitchen Recipe Scaling**. State recipe weights in grams (g) and times using the digital clock.

✎ Candidate Working Grid • Catering Kitchen Recipe Scaling

VOCATIONAL CALCULATION WORKSHOP • WORKSHOP 4

W4

Workshop 4: Warehouse Stock Inventory & Packing

Workplace Scenario

Counting boxes, dividing items into crates, and checking remaining stock.

Use your maths skills to complete the workplace tasks below:

1 **Task 1 (Crate Packaging):** A warehouse worker packs 48 coffee mugs into cardboard boxes. Each box holds 6 mugs. How many full boxes does he pack? **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Pallet Quantity):** A pallet holds 120 mugs. How many mugs remain unpacked after filling the 48-mug order? **(2 marks)**

Show your working

Answer: _____

CANDIDATE WORKSPACE • WORKSHOP 4

Workshop 4 • Warehouse Stock Inventory & Packing Working Space

Workspace Instructions • Warehouse Stock Inventory & Packing

Use this workspace to lay out calculation steps or diagram sketches for **Warehouse Stock Inventory & Packing**. State the full box count and the exact remainder count.

✎ Candidate Working Grid • Warehouse Stock Inventory & Packing

VOCATIONAL CALCULATION WORKSHOP • WORKSHOP 5

W5

Workshop 5: Building Maintenance & Fencing

Workplace Scenario

Measuring metres, calculating perimeter, and counting timber posts.

Use your maths skills to complete the workplace tasks below:

1 **Task 1 (Perimeter Measurement):** A rectangular garden plot measures 10 metres long and 6 metres wide. Calculate the total perimeter to be fenced. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Post Count):** A wooden post is placed every 2 metres along the 32-metre perimeter. How many posts are needed in total? **(2 marks)**

Show your working

Answer: _____

CANDIDATE WORKSPACE • WORKSHOP 5

Workshop 5 • Building Maintenance & Fencing Working Space

Workspace Instructions • Building Maintenance & Fencing

Use this workspace to lay out calculation steps or diagram sketches for **Building Maintenance & Fencing**. Ensure perimeter calculations state metres (m) and post counts are exact.

✎ Candidate Working Grid • Building Maintenance & Fencing

VOCATIONAL CALCULATION WORKSHOP • WORKSHOP 6

W6 Workshop 6: Logistics Driver Delivery Schedules

Workplace Scenario

Reading digital time, estimating journey durations, and tracking compass directions.

Use your maths skills to complete the workplace tasks below:

1 **Task 1 (Journey Duration):** A delivery courier departs depot at 09:15 and arrives at the first delivery address at 10:00. How many minutes did the journey take? **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Compass Heading):** The driver travels North along the high street, then makes a 90° clockwise turn. Which compass direction is the van now heading? **(2 marks)**

Show your working

Answer: _____

CANDIDATE WORKSPACE • WORKSHOP 6

Workshop 6 • Logistics Driver Delivery Schedules Working Space

Workspace Instructions • Logistics Driver Delivery Schedules

Use this workspace to lay out calculation steps or diagram sketches for **Logistics Driver Delivery Schedules**. Record journey duration in minutes and specify compass travel directions.

✎ Candidate Working Grid • Logistics Driver Delivery Schedules

VOCATIONAL CALCULATION WORKSHOP • WORKSHOP 7

W7

Workshop 7: Office Supply Ordering & Budgeting

Workplace Scenario

Adding equipment costs, rounding to nearest £10, and verifying receipts.

Use your maths skills to complete the workplace tasks below:

1 **Task 1 (Stationery Budget):** An office manager buys 5 reams of paper at £4.50 each and 2 black ink cartridges at £14.00 each. Find the total order cost. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Estimation):** Round the order total (£50.50) to the nearest £10 to give a quick budget estimate. **(2 marks)**

Show your working

Answer: _____

CANDIDATE WORKSPACE • WORKSHOP 7

Workshop 7 • Office Supply Ordering & Budgeting Working Space

Workspace Instructions • Office Supply Ordering & Budgeting

Use this workspace to lay out calculation steps or diagram sketches for **Office Supply Ordering & Budgeting**. State rounded order totals to the nearest £10.

✎ Candidate Working Grid • Office Supply Ordering & Budgeting

VOCATIONAL CALCULATION WORKSHOP • WORKSHOP 8

W8

Workshop 8: Community Centre Event Planning

Workplace Scenario

Sorting attendee data, creating tally charts, and drawing bar graphs.

Use your maths skills to complete the workplace tasks below:

1 **Task 1 (Attendee Count):** A community event registers 24 adults and 18 children. Calculate the total number of attendees. **(2 marks)**

Show your working

Answer: _____

2 **Task 2 (Chair Audit):** The hall has 45 chairs available. Are there enough chairs for everyone? How many chairs are spare or short? **(2 marks)**

Show your working

Answer: _____

CANDIDATE WORKSPACE • WORKSHOP 8

Workshop 8 • Community Centre Event Planning Working Space

Workspace Instructions • Community Centre Event Planning

Use this workspace to lay out calculation steps or diagram sketches for **Community Centre Event Planning**. Double-check attendee counts and verify whether seating capacity is sufficient.

✎ Candidate Working Grid • Community Centre Event Planning

ENTRY 3 EXAM PRACTICE • SECTION A • SHEET 1 • NO CALCULATOR

M1A

Mock Exam 1 • Section A (Non-Calculator)

1 **Question 1:** Calculate: **487 + 265** using formal column addition. **(2 marks)**

Show your working

Answer: _____

2 **Question 2:** Calculate: **632 – 268** using column subtraction, and write down an addition check. **(2 marks)**

Show your working

Answer: _____

ENTRY 3 EXAM PRACTICE • SECTION A • SHEET 2 • NO CALCULATOR

M1A

Mock Exam 1 • Section A (Non-Calculator)

3 **Question 3:** A team has 36 members. ¾ of the team attend a health and safety briefing. How many members attend the briefing? **(2 marks)**

Show your working

Answer: _____

4 **Question 4:** A customer buys a sandwich for £3.45 and a coffee for £2.80. She pays with a £10 note. How much change does she receive? **(2 marks)**

Show your working

Answer: _____

ENTRY 3 EXAM PRACTICE • SECTION A • SHEET 3 • NO CALCULATOR

M1A

Mock Exam 1 • Section A (Non-Calculator)

5 **Question 5:** A bus journey starts at 08:35 and arrives at 09:20. How many minutes did the journey take? **(2 marks)**

Show your working

Answer: _____

6 **Question 6:** A rectangular storeroom measures 8 metres long and 5 metres wide. Calculate the perimeter of the storeroom in metres. **(2 marks)**

Show your working

Answer: _____

ENTRY 3 EXAM PRACTICE • SECTION B • SHEET 1 • CALCULATOR PERMITTED

M1B

Mock Exam 1 • Section B (Calculator Permitted)

7 **Question 7:** A company orders 8 office chairs at £45 each and a desk for £185. What is the total cost of the order? **(3 marks)**

Show your working

Answer: _____

8 **Question 8:** A night shift begins at 21:30 and finishes at 06:00 the next morning. How many hours and minutes does the shift last? **(2 marks)**

Show your working

Answer: _____

ENTRY 3 EXAM PRACTICE • SECTION B • SHEET 2 • CALCULATOR PERMITTED

M1B

Mock Exam 1 • Section B (Calculator Permitted)

9 **Question 9:** A factory packs 92 mugs into gift boxes. Each box holds 6 mugs. How many full boxes can be packed, and how many mugs are left over? **(2 marks)**

Show your working

Answer: _____

10 **Question 10:** A chef needs 1.5 kg of flour. He has three bags containing 450g, 550g, and 350g. Does he have enough flour in total? Show calculations to justify your answer. **(3 marks)**

Show your working

Answer: _____

ENTRY 3 EXAM PRACTICE • SECTION B • SHEET 3 • CALCULATOR PERMITTED

M1B

Mock Exam 1 • Section B (Calculator Permitted)

11 **Question 11:** A bottle of liquid medicine contains 300 ml. A patient takes a 15 ml dose three times a day. How many days will the bottle last? **(2 marks)**

Show your working

Answer: _____

12 **Question 12:** At 06:00, the temperature in a cold storage unit was −15°C. By 14:00, the temperature had risen by 8°C. What was the temperature at 14:00? **(2 marks)**

Show your working

Answer: _____

ENTRY 3 EXAM PRACTICE • SECTION B • SHEET 4 • CALCULATOR PERMITTED

M1B

Mock Exam 1 • Section B (Calculator Permitted)

13 **Question 13:** A pictogram records bicycle rentals where each ★ represents 8 bicycles. **(3 marks)**

Monday has 3 full stars.

Tuesday has 2 and a half stars.

How many bicycles were rented in total over the two days?

Show your working

Answer: _____

14 **Question 14:** A delivery drone travels North from depot A to depot B, then turns 90° clockwise. Which compass direction is the drone now travelling? **(2 marks)**

Show your working

Answer: _____

ENTRY 3 EXAM PRACTICE • SECTION B • SHEET 5 • CALCULATOR PERMITTED

M1B

Mock Exam 1 • Section B (Calculator Permitted)

15 **Question 15:** A bar chart shows staff absences across four quarters: Q1: 15, Q2: 25, Q3: 10, Q4: 30. What was the total number of absences for the whole year, and what is the difference between Q4 and Q3? **(3 marks)**

Show your working

Answer: _____

16 **Question 16:** A survey of 40 apprentices records commute method: 18 travel by Train, 14 by Bus, and the rest Walk. How many apprentices walk to work? **(2 marks)**

Show your working

Answer: _____

ENTRY 3 EXAM PRACTICE • SECTION A • SHEET 1 • NO CALCULATOR

M2A

Mock Exam 2 • Section A (Non-Calculator)

1 **Question 1:** A logistics depot logs inbound deliveries: calculate **245 + 168** using columnar addition. **(2 marks)**

Show your working

Answer: _____

2 **Question 2:** An apprentice checks stock levels: evaluate **500 – 245** and show an addition check to verify your answer. **(2 marks)**

Show your working

Answer: _____

ENTRY 3 EXAM PRACTICE • SECTION A • SHEET 2 • NO CALCULATOR

M2A

Mock Exam 2 • Section A (Non-Calculator)

3 **Question 3:** A florist prepares 6 identical bouquets. Each bouquet contains 4 red roses. How many red roses are used in total? **(2 marks)**

Show your working

Answer: _____

4 **Question 4:** A timber strip is 40 cm long. A carpenter cuts off ¼ of the strip. What length of timber was cut off? **(2 marks)**

Show your working

Answer: _____

ENTRY 3 EXAM PRACTICE • SECTION A • SHEET 3 • NO CALCULATOR

M2A

Mock Exam 2 • Section A (Non-Calculator)

5 **Question 5:** A customer buys stationery: a notebook for £4.20, a pen set for £3.50, and folders for £6.10. She pays with a £20 note. How much change does she receive? **(3 marks)**

Show your working

Answer: _____

6 **Question 6:** A rectangular office room measures 8 metres long and 5 metres wide. Calculate the perimeter of the room in metres. **(2 marks)**

Show your working

Answer: _____

ENTRY 3 EXAM PRACTICE • SECTION B • SHEET 1 • CALCULATOR PERMITTED

M2B

Mock Exam 2 • Section B (Calculator Permitted)

7 **Question 7:** An afternoon appointment is scheduled for 3:30 pm. Write this time using the 24-hour digital clock. **(2 marks)**

Show your working

Answer: _____

8 **Question 8:** An office manager orders 3 packs of printer paper at £5.50 per pack. Calculate the total cost. **(3 marks)**

Show your working

Answer: _____

ENTRY 3 EXAM PRACTICE • SECTION B • SHEET 2 • CALCULATOR PERMITTED

M2B

Mock Exam 2 • Section B (Calculator Permitted)

9 **Question 9:** A parcel weighs 1.2 kg. Convert this weight into grams. **(2 marks)**

Show your working

Answer: _____

10 **Question 10:** The morning temperature outside a warehouse was 12°C. In the afternoon, the temperature reached 18°C. What was the temperature difference? **(2 marks)**

Show your working

Answer: _____

ENTRY 3 EXAM PRACTICE • SECTION B • SHEET 3 • CALCULATOR PERMITTED

M2B

Mock Exam 2 • Section B (Calculator Permitted)

11 **Question 11:** A bar chart shows visitors to a garden centre: Saturday had 65 visitors and Sunday had 50 visitors. What is the difference between Saturday and Sunday? **(3 marks)**

Show your working

Answer: _____

12 **Question 12:** On a grid map of a retail park, the DIY Store is located at Column B, Row 3. Write down the grid reference for the DIY Store. **(2 marks)**

Show your working

Answer: _____

ENTRY 3 EXAM PRACTICE • SECTION B • SHEET 4 • CALCULATOR PERMITTED

M2B

Mock Exam 2 • Section B (Calculator Permitted)

13 **Question 13:** A learner rolls a standard 6-sided dice numbered 1 to 6. Choose the probability word that describes rolling a 7: Certain, Likely, Unlikely, or Impossible. **(2 marks)**

Show your working

Answer: _____

14 **Question 14:** A workshop apprentice faces North, then turns 90° clockwise. Which compass direction is the apprentice now facing? **(2 marks)**

Show your working

Answer: _____

ENTRY 3 EXAM PRACTICE • SECTION B • SHEET 5 • CALCULATOR PERMITTED

M2B

Mock Exam 2 • Section B (Calculator Permitted)

15 **Question 15:** A cafe audits hot drink orders on Friday: 35 coffees, 20 teas, and 15 hot chocolates. How many hot drinks were ordered in total? **(3 marks)**

Show your working

Answer: _____

16 **Question 16:** Round 178 to the nearest 10, and round 178 to the nearest 100. **(3 marks)**

Show your working

Answer: _____

REFERENCE SECTION: SOLUTIONS, GLOSSARY & INDEX

Complete Step-by-Step Worked Solutions & Statutory Guidance

TOPIC 1: PLACE VALUE UP TO 1,000

Task 1: 458; **Task 2:** 602; **Task 3:** Order: 148, 235, 420, 680 [6m].

TOPIC 2: COMPARING & ORDERING NUMBERS

Task 1: 564 > 546; **Task 2:** Lowest = 125, Highest = 480; **Task 3:** Ordered list [6m].

TOPIC 3: ADDITION: COLUMN METHOD

Task 1: 245 + 138 = **383**; **Task 2:** £34.50 + £18.20 = **£52.70**; **Task 3:** Total patients = 124 + 98 = **222** [6m].

TOPIC 4: SUBTRACTION WITH BORROWING

Task 1: 642 - 278 = **364**; **Task 2:** Stock = 450 - 185 = **265 boxes**; **Task 3:** Difference = 512 - 349 = **163** [6m].

TOPIC 5: TIMES TABLES (2, 3, 4, 5, 10)

Task 1: 8 × 5 = **40**; **Task 2:** 6 × 4 = **24 badges**; **Task 3:** 7 packs of 10 = **70 items** [6m].

TOPIC 6: MULTIPLYING 2-DIGIT BY 1-DIGIT

Task 1: 24 × 3 = **72**; **Task 2:** 35 × 4 = **140 items**; **Task 3:** 42 × 5 = **210 units** [6m].

TOPIC 7: DIVISION & REMAINDERS

Task 1: 28 ÷ 4 = **7**; **Task 2:** 29 ÷ 4 = **7 each, 1 leftover**; **Task 3:** 45 ÷ 5 = **9 teams** [6m].

TOPIC 8: FRACTIONS: 1/2, 1/4, 1/3, 1/10

Task 1: Half of £40 = **£20.00**; **Task 2:** Quarter of 32 m = **8 metres**; **Task 3:** One-tenth of 100 kg = **10 kg** [6m].

TOPIC 9: DECIMALS & MONEY NOTATION

Task 1: Seventy-five pence = **£0.75**; **Task 2:** £4.50 + £2.25 = **£6.75**; **Task 3:** £10.00 - £6.50 = **£3.50** [6m].

TOPIC 10: ROUNDING TO NEAREST 10 & 100

Task 1: 67 to nearest 10 = **70**; **Task 2:** 235 to nearest 100 = **200**; **Task 3:** 85 rounds UP to **90** [6m].

TOPIC 11: ADDING COSTS & GIVING CHANGE

Task 1: Spend: £2.85 + £1.65 = £4.50. Change from £5 = **£0.50** (50p); **Task 2:** Change from £10 for £6.40 = **£3.60**; **Task 3:** £20 - £14.50 = **£5.50** [6m].

TOPIC 12: TELLING TIME (12H & 24H)

Task 1: Quarter past 9 am = **9:15 am**; **Task 2:** 2:00 pm in 24h = **14:00**; **Task 3:** Half past 4 pm = **16:30** [6m].

TOPIC 13: MEASURING LENGTH (MM, CM, M)

Task 1: 1 m 40 cm = **140 cm**; **Task 2:** 25 mm = **2.5 cm**; **Task 3:** Total length = 3.2 m = **320 cm** [6m].

TOPIC 14: MEASURING WEIGHT (G AND KG)

Task 1: Half a kg = **500 grams**; **Task 2:** 1.5 kg = **1,500 g**; **Task 3:** Parcel = 750 g + 250 g = **1 kg** [6m].

REFERENCE SECTION: SOLUTIONS, GLOSSARY & INDEX (CONTINUED)

WORKSHOP 3: CATERING KITCHEN RECIPES

T1: Double recipe: 12 muffins need 250 g → 24 muffins need **500 g flour**. **T2:** Baking time: 25 minutes from 10:15 → **10:40**.

WORKSHOP 4: WAREHOUSE STOCK PACKING

T1: 48 mugs packed 6 to a box = 48 ÷ 6 = **8 boxes**. **T2:** Remaining stock = 120 - 48 = **72 mugs**.

WORKSHOP 5: BUILDING MAINTENANCE & FENCING

T1: Perimeter = 2(10 + 6) = **32 metres**. **T2:** Posts needed: 1 post every 2 metres = 32 ÷ 2 = **16 posts**.

WORKSHOP 6: DRIVER DELIVERY SCHEDULES

T1: Journey: departs 09:15, travel 45 mins → arrives **10:00**. **T2:** Head North, turn right → facing **East**.

WORKSHOP 7: OFFICE SUPPLY ORDERING

T1: 5 boxes of paper at £4.50 = **£22.50**. **T2:** Round £22.50 to nearest £10 = **£20.00**.

WORKSHOP 8: COMMUNITY EVENT PLANNING

T1: Tally count: Adults 24, Children 18 → Total = **42 attendees**. **T2:** Chairs needed: 45 chairs available → **Yes, sufficient chairs**.

OFFICIAL MOCK EXAM 1 (36 MARKS)

Section A (Non-Calculator, 9 Marks):
Q1: 158 + 236 = **394** [2m]. **Q2:** 450 - 185 = **265** [2m]. **Q3:** 8 × 5 = **40** [2m]. **Q4:** Half of £36 = **£18.00** [3m].
Section B (Calculator Permitted, 27 Marks):
Q5: Total bill = £3.45 + £2.80 + £5.25 = £11.50. Change from £20 = **£8.50** [3m].
Q6: Perimeter of garden 9 m × 6 m = 2(9 + 6) = **30 metres** [3m].
Q7: Time: 09:30 to 11:15 = **1 hour 45 minutes** [3m].
Q8: 4 boxes of tiles at £14.50 = **£58.00** [3m].
Q9: Liquid: 2.5 litres = **2,500 ml** [3m].
Q10: Tally chart frequency for Car = **14** [3m].
Q11: Bar chart: Most popular sport is Football with **28 votes** [3m].
Q12: Probability of getting tails on a coin is **Evens** [3m].
Q13: Round 178 to nearest 100 = **200** [3m].

OFFICIAL MOCK EXAM 2 (36 MARKS)

Section A (Non-Calculator, 9 Marks):
Q1: 245 + 168 = **413** [2m]. **Q2:** 500 - 245 = **255** [2m]. **Q3:** 6 × 4 = **24** [2m]. **Q4:** Quarter of 40 cm = **10 cm** [3m].
Section B (Calculator Permitted, 27 Marks):
Q5: Cost: £4.20 + £3.50 + £6.10 = £13.80. Change from £20 = **£6.20** [3m].
Q6: Perimeter = 2(8 + 5) = **26 metres** [3m].
Q7: Digital 24h: 3:30 pm = **15:30** [3m].
Q8: 3 packs of paper at £5.50 = **£16.50** [3m].
Q9: 1.2 kg = **1,200 grams** [3m].
Q10: Difference in temperature: 18°C - 12°C = **6°C** [3m].
Q11: Bar chart: Difference between Saturday and Sunday = **15 customers** [3m].
Q12: Grid reference for Store = **B3** [3m].
Q13: Likelihood of rolling a 7 on a 6-sided die is **Impossible** [3m].

ENTRY LEVEL 3 KEY RULES & CONVERSIONS

Addition & Subtraction:

• Column addition: add units first, then tens, then hundreds (carry over into next column).

• Column subtraction: if top digit is smaller, borrow 1 from the column to the left.

Multiplication & Division:

• Learn 2, 3, 4, 5, and 10 times tables.

• Multiplying by 10 shifts digits one place to the left (with a trailing zero).

• Division shares into equal groups; leftover amount is the remainder.

Fractions & Money:

• 1/2 = half (divide by 2) 1/4 = quarter (halve and halve again)

• 100 pence = 1 pound (£) Change = Amount given − Total cost

Everyday Measures:

• 1 metre = 100 cm 1 cm = 10 mm

• 1 kilogram = 1,000 grams 1 litre = 1,000 millilitres

• Perimeter = add all 4 sides of a rectangle

REFERENCE SECTION: SOLUTIONS, GLOSSARY & INDEX (CONTINUED)

MATHEMATICAL GLOSSARY & VOCABULARY

Add
Combining two or more numbers together to find the total sum (+).

Bar Chart
A chart with rectangular bars showing amounts for different categories.

Borrowing
Taking 1 ten or 1 hundred from the left column in subtraction when the top digit is smaller.

Capacity
The amount of liquid a container can hold, measured in millilitres (ml) or litres (l).

Centimetre (cm)
A metric unit of length: 100 cm = 1 metre.

Change
The money returned to you when you pay with more than the bill amount.

Digit
Any single number symbol from 0 to 9 used to write numbers.

Division
Sharing an amount into equal groups or finding how many groups fit into a total (÷).

Gram (g)
A metric unit of mass/weight: 1,000 grams = 1 kilogram.

Half
One of two equal parts of a whole (1/2).

Kilogram (kg)
A metric unit of heavy weight: roughly the weight of a bag of sugar.

Litre (l)
A metric unit of liquid capacity: 1 litre = 1,000 ml.

Perimeter
The total distance all the way around the outside edge of a shape.

Quarter
One of four equal parts of a whole (1/4).

Remainder
The amount leftover when a number cannot be divided equally.

Subtract
Taking one number away from another to find the difference (-).

Tally
Counting in groups of five using vertical marks and a diagonal strike.

CURRICULUM & TOPIC INDEX

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EVAL

ENTRY LEVEL 3 • SELF-EVALUATION

Learner Competency Tracker & Revision Log

Entry Level 3 Core Mathematical Standards

Track your confidence across the core Entry Level 3 Functional Skills mathematical standards. Tick each topic as you master it:

- **Number Operations:** Addition & subtraction with regrouping/borrowing; times tables (2, 3, 4, 5, 10); sharing with remainders.
- **Fractions & Decimals:** Finding 1/2, 1/4, 1/3, 1/10 of amounts; decimals in everyday money (£ and pence).
- **Measures & Shapes:** Length (mm, cm, m), weight (g, kg), capacity (ml, l); telling 12h/24h time; calculating perimeter.
- **Handling Information:** Extracting data from tally charts, lists, and bar charts; using grid references and compass directions.

✎ Personal Revision Targets & Tutor Diagnostic Action Plan

AWARD

ENTRY LEVEL 3 • WELL DONE!

Certificate of Progress & Achievement

Functional Skills Mathematics Entry Level 3 Competencies

Well done on completing the units and practice papers in this study manual! You have mastered:

- Counting, ordering, adding, and subtracting numbers up to 1,000.
- Using times tables (2, 3, 4, 5, 10) and sharing quantities.
- Working with halves, quarters, tenths, and decimal money.
- Measuring length, mass, and capacity with metric units.
- Reading clock faces, 12h/24h digital time, and temperature scales.
- Reading data from tally charts, tables, and bar charts.

✎ Tutor Endorsement & Candidate Next Steps

Build solid numeracy foundations and achieve Entry Level 3

Engineered for adult learners, apprentices, and ESOL candidates building foundational math skills. Takes you step-by-step through whole numbers up to 1000, addition, subtraction, multiplication grids, division remainders, decimal money, and basic clock time.

INSIDE THIS PUBLICATION

- ✔ **Dyslexia-Friendly Layout:** High-contrast visual models, chunked explanations, and generous font spacing.
- ✔ **Whole Numbers & Place Value:** Clear visual place value charts (Hundreds, Tens, Units) and number line jumps.
- ✔ **Everyday Practical Arithmetic:** Real-world calculations: shopping change, bus timetables, and measuring recipes.
- ✔ **Illustrated Step-by-Step Models:** Step-by-step breakdowns showing learners exactly how to complete each calculation.
- ✔ **Confidence-Building Checkpoints:** Self-check progress grids at the end of each unit to verify mastery.

Progress to Functional Skills Level 1:

Once Entry Level 3 is mastered, learners transition directly to the *Functional Skills Maths Level 1 Core Study Guide* for accredited Level 1 qualification.

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